

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Weeks 37
21–27 September 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



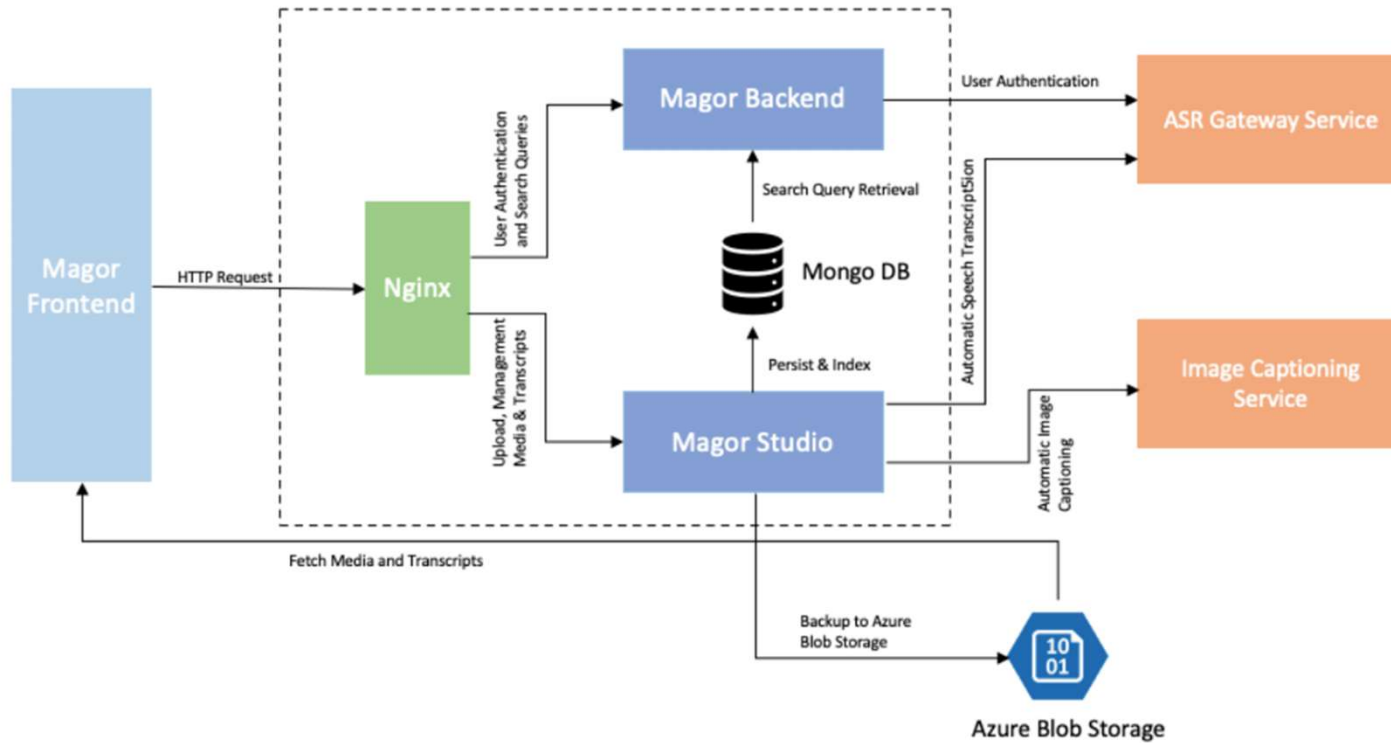
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work Done Over The Weeks:

- Updated Summary requests to use taskType: GENERAL and queue: meadow9
- Updated Q&A requests to use taskType: CHAT and queue: meadow9
 - Pass previous Q&A through the dedicated chatHistory attribute instead of embedding it in the prompt



Task 1 (Update Summary Request Configurations)

- Use taskType: GENERAL and queue: meadow9 :

```
[AI Summarize] Payload: {  
  "taskType": "GENERAL",  
  "queue": "meadow9",  
  "dialogue": "channel1_S1 00:00.11 00:03.52 So, everyone, welcome back to another episode of Deep Dive\nchannel1_S1 00:03.53 00:05.57 with me,  
Steven and Michael. Tiffany.\nchannel1_S1 00:06.05 00:09.04 Now, um, this week, we're talking about, uh, the political\nchannel1_S1 00:09.04  
00:11.43 scene in Singapore because the Workers Party leader, Pritam Singh,\nchannel1_S1 00:11.43 00:13.82 is no longer Singapore's leader of the  
opposition.\nchannel1_S2 00:09.63 00:09.94 Because ah\nchannel1_S2 00:14.55 00:17.88 Prime Minister Lawrence Wong, he has removed mr Singh  
from\nchannel1_S2 00:17.88 00:21.21 this role and invited the Workers' Party to nominate another\nchannel1_S2 00:21.21 00:23.05 person to take the  
seat.\nchannel1_S1 00:14.79 00:15.21 Right.\nchannel1_S1 00:19.39 00:19.65 Yeah.\nchannel1_S1 00:23.79 00:26.74 Yeah, that's right. And this is all  
coming on the\nchannel1_S1 00:26.74 00:29.69 back of Mr. Singh's court convictions for lying to the\nchannel1_S1 00:29.69 00:32.63 Committee of  
Privileges. Although, whether he had instructed former MP\nchannel1_S1 00:32.64 00:35.58 Ratio Khan to come clean about a lie she told\nchannel1_S1 00:35.59 00:36.17 in parliament.\nchannel1_S2 00:36.51 00:40.03 And last week, the house passed the motion to deem\nchannel1_S2 00:39.95 00:43.77  
Mr. Singh unfit to stand as leader of the opposition.\nchannel1_S1 00:39.79 00:40.03 Yeah.\nchannel1_S1 00:44.17 00:46.59 So what will the next  
move by the Workers Party\nchannel1_S1 00:46.59 00:49.00 be? You know, uh, what kind of impact will this\nchannel1_S1 00:49.01 00:51.42 have on  
them? Well, to help us with our discussion\nchannel1_S1 00:51.42 00:52.63 today, we have with us\nchannel1_S1 00:52.79 00:55.97 Associate Professor  
Eugene Tan from the Singapore Management University\nchannel1_S3 00:56.21 00:58.09 Hi, Steve. Hi, Tiffany. Thank you for having me.\nchannel1_S1 00:58.33 01:00.97 An independent political observer, dr Felix Tan.\nchannel1_S3 01:01.11 01:01.79 Hi.\nchannel1_S1 01:01.79 01:03.37 All of you,  
okay? Welcome guys.\nchannel1_S2 01:08.47 01:11.58 Well, the Workers Party said that it will deliberate on\nchannel1_S2 01:11.58 01:14.69 PM's  
letter and will respond to the invitation in due\nchannel1_S2 01:14.69 01:17.79 course on whether they will take up the leader of\nchannel1_S2 01:17.79 01:20.90 the opposition replacement role. I mean, basically, nominate someone to\nchannel1_S2 01:20.90 01:24.01 take over the, the seat.  
But I guess the announcement\nchannel1_S2 01:24.01 01:27.12 to remove Mister Singh really, you know, didn't come as\nchannel1_S2 01:27.12 01:30.22 a, as a big surprise. Correct. Erm, given that Mister\nchannel1_S2 01:30.22 01:33.33 Singh is also the WP Sec Gen, will this cause\nchannel1_S2 01:33.33 01:36.44 a leadership crisis within the party now that they may\nchannel1_S2 01:36.44 01:39.54 have to, not, they will have to nominate a  
leader of the opposition if they take the invitation? Not to worry.\nchannel1_S2 01:43.65 01:44.07 I see the party is
```



Task 2 (Update Q&A Request Configurations)

- Updated Q&A request:

```
2026-09-24T13:49:30.130Z - info: GET /api/ai/qa-task-status?mediaId=6aabec952567e700242aa707 304 2.002 ms - - [morgan]
[AI QA] Sending request to SpeechLab gateway:
[AI QA] URL: https://gateway.speechlab.sg/llm
[AI QA] Payload: {
  "taskType": "CHAT",
  "queue": "meadow9",
  "chatHistory": [
    {
      "role": "user",
      "content": "what is this about?"
    },
    {
      "role": "assistant",
      "content": "This transcript discusses speculation and media reports surrounding Chinese President Xi Jinping's health and conduct during his visit to New Delhi for the BRICS summit. It highlights concerns based on footage showing Xi appearing unsteady while disembarking, briefly pausing during an event, and not attending the summit's official dinner. Some sources suggest possible health issues, including a recurrence of past stroke symptoms, while others speculate about diplomatic tensions, such as perceived low reception by India or disputes involving translators, which may have led to his absence from the dinner. The report also touches on geopolitical implications, including U.S. pressure on China regarding Taiwan. Overall, it reflects international scrutiny and speculation about Xi's health and diplomatic standing."
    },
    {
      "role": "user",
      "content": "is this language chinese?"
    }
  ]
}
```

taskType: CHAT & queue: meadow9

chatHistory with role-content objects (no longer in dialogue below)

Removed userPrompt (was used to set Q&A response format) use gateway instructions



Task 2 (Implemented Q&A history pagination)

- Q&A exchanges after change:

Summarize Q&A

Ask a question about the transcript...

Ask

New Response Format (may vary)

- **MSCI:** Global index provider warning Indonesia may be downgraded from emerging to frontier market status if reforms aren't made by May 2026.
- **Bank Indonesia:** Indonesia's central bank, facing scrutiny over its independence due to the nomination of the president's nephew as deputy governor.
- **President Prabowo Subianto:** His nephew, Thomas, was nominated for a top central bank position, sparking debate over political influence.
- **Foreign investors:** Expressing concerns over transparency, ownership concentration, and fiscal discipline.
- **CNA (Channel News Asia) reporter Chani:** Reporting live from Jakarta on the developments.

Q: what is this about?

A: This is about the sharp decline in Indonesian shares, with the benchmark index plunging nearly 9% and triggering a circuit breaker, followed by concerns over market investability raised by MSCI (00:00–00:07). MSCI has warned of a possible downgrade to frontier market status if Indonesia does not address issues like thin trading, ownership concentration among wealthy families, and lack of transparency by May 2026 (00:11–00:26, 02:50–03:00). Additional concerns include the independence of Bank Indonesia, highlighted by the nomination of the president's nephew as deputy governor, and broader fiscal discipline issues following surprise rate cuts and the resignation of the finance minister (00:51–01:06, 02:04–02:20).

Previous Response format (userPrompt)

Summarize Q&A

Ask a question about the transcript...

Ask

Previous Q&A

Q: is this language chinese?

A: Yes, the primary language of the transcript is Mandarin Chinese, with some mixed-in English phrases and fragmented speech (e.g., "Is it okay?", "you room", "Betul"). The content appears to be from a news or commentary video, likely from a Chinese-speaking media outlet, discussing international political events involving Chinese President Xi Jinping.

Q: what is this about?

A: This transcript discusses speculation and media reports surrounding Chinese President Xi Jinping's health and conduct during his visit to New Delhi for the BRICS summit. It highlights concerns based on footage showing Xi appearing unsteady while disembarking, briefly pausing during an event, and not attending the summit's official dinner. Some sources suggest possible health issues, including a recurrence of past stroke symptoms, while others speculate about diplomatic tensions, such as perceived low reception by India or disputes involving translators, which may have led to his absence from the dinner. The report also touches on geopolitical implications, including U.S. pressure on China regarding Taiwan. Overall, it reflects international scrutiny and speculation about Xi's health and diplomatic standing.



Plans for Next Update

- Update FYP report and presentation slides
- Address any bugs or new tasks

